

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project

Analysis of 3,900+ customer shopping records to identify purchasing behavior, customer segments, revenue trends, and actionable business insights using Python, SQL, and Power BI.

Tools Used

- Python (Pandas)
- SQL (MySQL)
- Power BI
- GitHub

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Data Analytics Portfolio

GitHub Repository:

github.com/qevinattaqwarx-ui/customer-shopping-behavior-analysis

Project Overview

Business Problem

Businesses need to understand customer purchasing behavior, product performance, and revenue trends to make informed decisions, optimize marketing strategies, and improve overall business performance.

Project Objective

This project was conducted to:

- Analyze customer shopping behavior and purchasing patterns.
- Identify customer segments and demographic trends.
- Evaluate revenue contribution across different customer groups.
- Generate actionable business insights through data analysis and visualization.
- Develop an interactive dashboard to support data-driven decision-making.

Dataset Information

Dataset Size: 3,900 Customer Records

Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Product Category, Item Purchased, Purchase Amount, Quantity)
- Shopping behavior (Discount Usage, Previous Purchases, Purchase Frequency)
- Customer feedback (Review Rating)

Project Scope

This end-to-end analytics project covers:

- ✓ Data Cleaning & Standardization (Python)
- ✓ Exploratory Data Analysis (SQL)
- ✓ Business Insight Generation
- ✓ Interactive Dashboard Development (Power BI)
- ✓ Data Visualization & Reporting

Analytics Workflow

Raw Data
↓
Python Data Cleaning & Standardization
↓
SQL Exploratory Data Analysis
↓
Business Insights
↓
Power BI Dashboard

Data Analysis Process

Python Data Cleaning & Preparation

Python (Pandas) was used to prepare the dataset before analysis.

Key activities:

- Loaded and explored the dataset structure.
- Checked data quality using information and statistical summaries.
- Handled missing values in the Review Rating column using category-based median imputation.
- Standardized column names into snake_case format.
- Created additional features:
 - age_group
 - purchase_frequency_days
- Removed redundant features after consistency validation.
- Loaded the cleaned dataset into MySQL for further analysis.

Initial Data Exploration: df.info() before cleaning

Missing Value Handling

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customer_id           3900 non-null   int64
1   age                   3900 non-null   int64
2   gender                3900 non-null   str
3   item_purchased        3900 non-null   str
4   category              3900 non-null   str
5   purchase_amount       3900 non-null   int64
6   location              3900 non-null   str
7   size                  3900 non-null   str
8   color                 3900 non-null   str
9   season                3900 non-null   str
10  review_rating         3863 non-null   float64
11  subscription_status    3900 non-null   str
12  shipping_type          3900 non-null   str
13  discount_applied       3900 non-null   str
14  promo_code_used        3900 non-null   str
15  previous_purchases     3900 non-null   int64
16  payment_method         3900 non-null   str
17  frequency_of_purchases 3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

Customer ID	0
Age	0
Gender	0
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	0
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0
Payment Method	0
Frequency of Purchases	0
dtype: int64	

SQL Exploratory Data Analysis

SQL was used to answer business questions and identify customer behavior patterns.

Analysis performed:

- Revenue comparison by customer demographics.
- Discount effectiveness analysis.
- Product rating evaluation.
- Shipping type spending comparison.
- Subscription behavior analysis.
- Customer segmentation analysis.
- Product performance analysis.
- Repeat buyer behavior analysis.
- Revenue contribution by age group.

Power BI Dashboard

An interactive dashboard was developed to visualize:

- Customer purchasing behavior.
- Revenue trends.
- Customer segments.
- Product performance.
- Demographic analysis.

The dashboard enables stakeholders to explore customer insights and support data-driven decisions.

Key Insights & Business Recommendations

1. Strong Customer Loyalty Base

Finding

Customer segmentation identified:

- 3,116 Loyal Customers
- 701 Returning Customers
- 83 New Customers

Insight

The business has a strong customer retention foundation with a high proportion of repeat customers. However, the limited number of new customers indicates an opportunity to improve customer acquisition.

Recommendation

- Develop targeted campaigns to attract new customers.
- Strengthen loyalty strategies to maintain repeat purchasing behavior.

2. Promotions Drive Higher-Value Purchases

Finding

839 customers used discounts while spending above the average purchase amount.

Insight

Discount campaigns were not only attracting low-spending customers but also encouraging higher-value transactions.

Recommendation

- Optimize promotional campaigns based on customer value and purchasing behavior.
- Focus discounts on products and segments with higher revenue potential.

3. Balanced Revenue Across Customer Demographics

Finding

Young Adult customers generated the highest revenue contribution with 62,143 in sales, while other age groups showed similar contribution levels.

Insight

Revenue is distributed across multiple customer groups, indicating a diverse customer base.

Recommendation

- Maintain multi-segment marketing strategies.
- Personalize campaigns based on customer characteristics.

4. Subscription Program Optimization Opportunity

Finding

Non-subscribers generated 170,436 total revenue, higher than subscribers. Repeat buyers were also mostly non-subscribers (2,518 vs 958 customers).

Insight

Frequent purchasing behavior does not automatically lead to subscription adoption.

Recommendation

- Evaluate subscription benefits.
- Introduce stronger incentives to increase subscription adoption.

Business Questions Answered

- ✓ Which customer segments contribute the most value?
- ✓ How do discounts influence customer spending behavior?
- ✓ Which demographics contribute the highest revenue?
- ✓ Does subscription status impact purchasing behavior?

Business Impact

This project demonstrates the ability to transform raw customer transaction data into actionable insights through data preparation, exploratory analysis, and visualization.

The analysis supports:

- Customer retention strategy development.
- Marketing campaign optimization.
- Revenue performance evaluation.
- Data-driven decision-making.